

Group Project Report:
YouTube Trailer Engagement & Box Office Performance

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Problem Overview

Studios make major marketing decisions before a film is released, often with limited knowledge of how audiences will respond. To reduce this uncertainty, they invest heavily in pre-release promotion, with YouTube trailers serving as one of the most visible and measurable marketing tools. These trailers generate engagement data such as views, likes, comments, and audience sentiment, which can provide early signals of public attention and interest.

However, it is not always clear whether these digital engagement metrics translate into financial outcomes. A trailer may attract high view counts without leading to ticket sales, while stronger forms of interaction, such as likes or comments, may better capture genuine audience enthusiasm. This creates an important problem for studios and distributors, that although large amounts of online engagement data are available in real time, its value for forecasting box office performance remains uncertain.

This project examines how YouTube trailer engagement can be used to better understand audience reaction and support film marketing decisions. From a business perspective, understanding whether trailer engagement reliably signals box office performance could help studios gauge audience interest earlier and set more realistic revenue expectations before release. From an analytical perspective, the project explores a broader marketing question: whether online consumer engagement can serve as a useful indicator of audience interest and campaign performance. The main research question is: Can YouTube trailer engagement metrics predict box office performance?

Dataset Description

This project uses two publicly available datasets that capture different dimensions of movie performance. The first is a YouTube trailer and sentiment dataset containing 1,105 observations and 18 variables. Each row represents a film and includes identifying information such as movie name, trailer link, video ID, and year, along with film-related attributes such as genre, rating, director, writer, star, country, company, runtime, and budget. It also includes trailer audience response measures, most notably sentiment scores broken into positive, neutral, and negative reactions, as well as a favorability score and vote count. This makes the dataset useful for capturing both the basic characteristics of a film and the level and tone of audience engagement surrounding its trailer.

The second dataset is a box office performance dataset containing 5,000 observations and 13 variables on films released between 2000 and 2024. Each row includes the release group (movie title), year, worldwide box office revenue, domestic revenue, foreign revenue, and the percentage split between domestic and foreign earnings. It also contains descriptive film information such as genres, rating, vote count, original language, and production countries. Compared with the YouTube dataset, this file is more focused on commercial outcomes, especially the revenue measures needed to evaluate financial performance.

Together, these datasets provide both the digital audience response side and the financial performance side of the same problem. Using movie title and release year as identifiers, there are 626 films in common between the two datasets, creating a substantial matched sample for analysis. This combined data allows the project to examine whether patterns in trailer-related engagement and sentiment are associated with box office outcomes.

Overview and Data Pre-processing

To ensure data consistency and accuracy from two separate sources, the datasets required cleaning. Column names were standardized, missing values were handled by dropping rows, and sentiment scores were parsed and converted to proportions. Revenues and budgets were log-transformed to reduce right skewness, and genre was one-hot encoded to avoid false ordering between categories. Duplicated rows were identified, dropped, and printed in an updated dataset using `df.drop_duplicates()`. Rows with missing values were dropped rather than imputed because key predictors like budget are essential to the analysis. Filling them with averages or placeholder values could introduce artificial patterns that mislead the models. Additionally, the inner merge between the YouTube and box office datasets naturally removed movies that did not appear in both sources, ensuring that every row used in modelling had complete trailer engagement data paired with verified revenue figures. While this reduced the dataset to 626 rows, data quality was prioritized over quantity to ensure more reliable results.

The dataset was split 80/20 into training and testing subsets to evaluate each model's ability to generalize to unseen data. Four models were used: a Baseline Linear Regression using only YouTube features, a Full Linear Regression incorporating control variables (budget, runtime, and genre), a Decision Tree, and a Random Forest. The Baseline model served as a foundation to measure general engagement, while the Full model isolated YouTube's true contribution. The Decision Tree and Random Forest captured non-linear relationships that linear regression may miss. Model performance was evaluated using R^2 , Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE), and results were compared across all four models to assess the tradeoff between simplicity and predictive accuracy.

Feature Engineering

The feature engineering step transformed the raw movie data into variables that could be used in the model. The worldwide revenue and the budget were skewed, so the *np.log1p* function was utilized to create *log_revenue* and *log_budget*, making the data more consistent and minimizing the influence of extreme values. The YouTube sentiment scores were stored as a single dictionary string, making them unusable as model inputs. They were parsed into separate numeric columns for positive, neutral, and negative sentiment, then converted to proportions so they could be fairly compared across movies with different levels of engagement. The neutral sentiment was removed to avoid a strong correlation between audience response variables, and rows with missing percentage values were also removed. One-hot encoding was selected instead of label encoding for genre since label encoding uses scales like “Action = 0, Thriller = 11”, which is incorrect since one type of movie cannot be more than another. One-hot encoding prevents this from occurring since a separate column is created for a genre. However, a drawback is that more columns are created in a dataset, which may affect model performance if there are fewer rows.

The feature engineering was performed in Google Colab using pandas, numpy, and matplotlib. The code first creates *log_revenue* and *log_budget* columns using *np.log1p*, then transforms the YouTube sentiment string into separate numeric columns and converts them to percentage values, and finally one-hot encodes the genre column. Each cell displays its output immediately below, including histograms confirming the log transformation reduced skewness, the updated dataframe shape after sentiment parsing, a preview of the new sentiment percentages, and a list of the newly created genre columns.

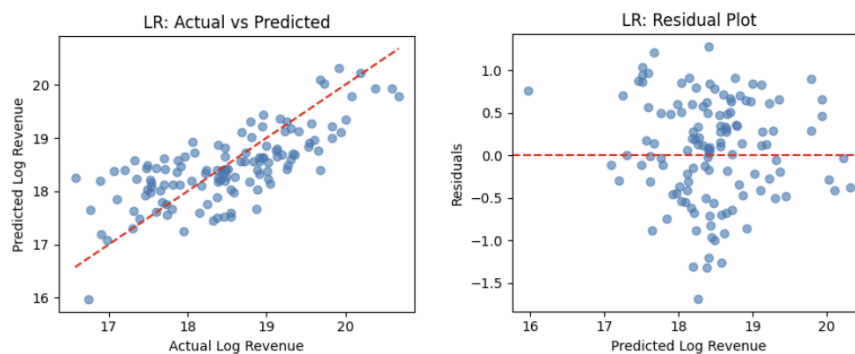
The feature engineering step was successful in data cleaning for modelling. The output indicates that the *worldwide_revenue* and budget columns were successfully transformed into *log_revenue* and *log_budget* columns, and the histogram indicates that *log_revenue* is not skewed, unlike the revenue column. The output from the sentiment splitting also indicates the shape of the data (624,16) after this step, and the preview suggests that the YouTube sentiment field was converted into useful percentage variables. For instance, the movie, *Harry Potter and the Goblet of Fire*, now has *_sent_positive_pct=0.348* and *sent_negative_pct=0.204*, whereas *Batman Begins*, now has *sent_positive_pct=0.512* and *sent_negative_pct=0.100*.

Baseline + Full Linear Regression

Linear regression was chosen as the baseline model because it provides a starting point for understanding which features drive box office revenue. Two versions were built: a baseline model using only YouTube trailer features to isolate how much trailer engagement alone explains revenue, and a full model that adds budget, runtime, and genre as control variables to isolate YouTube's impact from other variables that already influence revenue on their own. The trade-off for linear regressions assumes a straight-line relationship between features and revenue, which may oversimplify the true patterns in the data and limit prediction accuracy for films with unusual combinations. The target variable was log-transformed to reduce skew and improve model performance, but the predictions are converted back to actual dollar amounts through *np.expml* when reporting errors like MAE, so the results are easier to interpret. Features were standardized using *StandardScaler* so that variables with very different scales, such as budget versus sentiment percentages, could be fairly compared by their relative importance rather than their magnitude.

In order to run the linear regression, the libraries that are required for this include *pandas*, *numpy*, *sklearn.linear_model*, *train_test_split* and *matplotlib*. The baseline model cell fits on *YOUTUBE_FEATURES* only, while the full model cell uses all features, including controls.

The baseline linear regression model using only YouTube features achieved an R^2 of 0.3324, an MAE of \$81,530,126, and an RMSE of 0.7224, meaning trailer engagement alone explains roughly 33% of revenue variance. Adding budget, runtime, and genre as controls in the full model nearly doubled performance, raising R^2 to 0.526, and reducing MAE to \$69,090,603 and RMSE to 0.6087. The standardized coefficients show that *log_budget* (0.418) and *vote_count* (0.411) are the two strongest predictors, contributing nearly equally, while *youtube_favorability* (0.086) and *sent_negative_pct* (0.020) have smaller positive effects. The actual vs. predicted plot shows predictions clustering well around the diagonal for mid-range revenue films, while the residual plot confirms residuals are roughly centred at zero but spread out at higher predicted values.



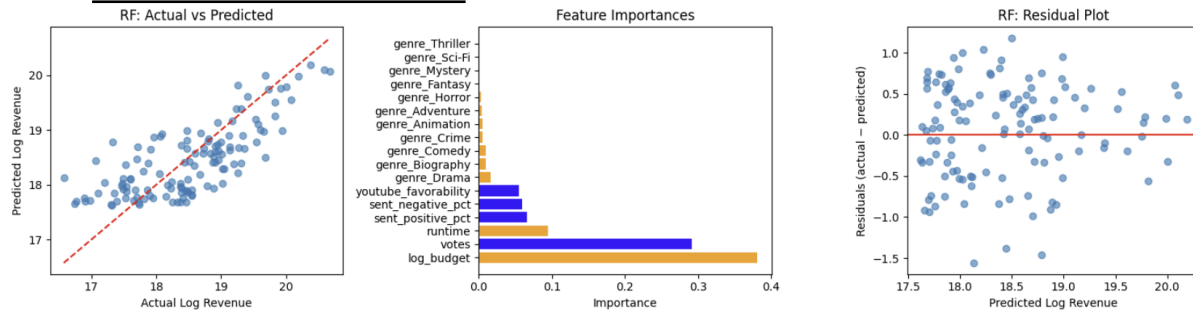
Decision Tree / Feature Importance (Top 10 Predictors)

A Decision Tree and a Random Forest were both built to capture non-linear relationships between features and revenue that linear regression cannot detect. The Decision Tree was capped at a maximum depth of 5 to prevent overfitting; without this limit, the tree would memorize the training data and perform poorly on unseen films. Random Forest was chosen as the stronger model because it builds hundreds of decision trees and averages their predictions, making it more stable and accurate than a single tree. The trade-off arises since though the Decision Tree is easy to interpret and visualize, Random Forest is harder to explain, even though it produces better results.

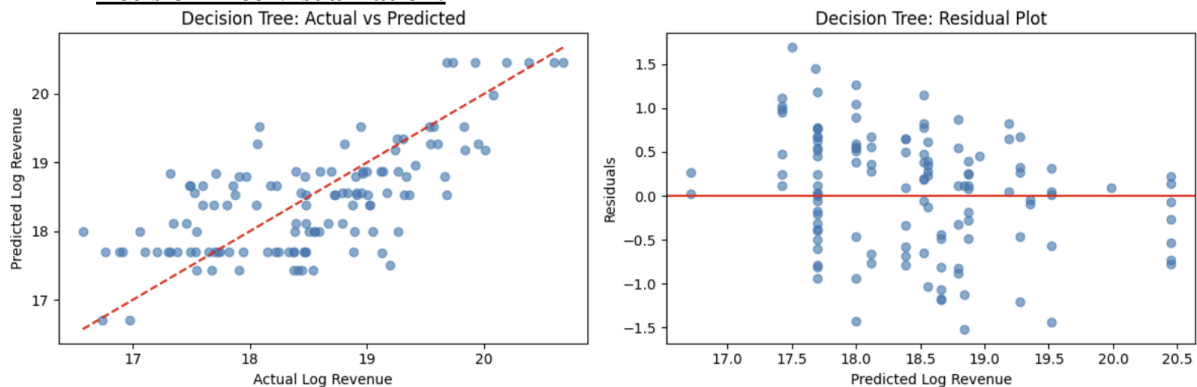
Both the Decision Tree and Random Forest cells depend on the same cleaned dataset and train/test split used throughout the notebook. The libraries that are required for this include *sklearn.tree.DecisionTreeRegressor* and *sklearn.ensemble.RandomForestRegressor* are both included in the imports cell at the top. All R^2 , MAE, and RMSE results print inline beneath each cell with no separate output files required.

The Decision Tree achieved an R^2 of 0.4319, an MAE of \$70,594,505, and an RMSE of 0.6664, while the Random Forest outperformed it with an R^2 of 0.573, an MAE of \$62,551,849, and an RMSE of 0.5777, both surpassing the YouTube-only baseline (R^2 of 0.3324) and confirming that non-linear models better capture the complexity of revenue prediction. The Random Forest feature-importance-rankings reveal that *log_budget* (0.381) and YouTube *vote_count* (0.291) are the two strongest predictors, together accounting for over 67% of the model's predictive power. YouTube sentiment features such as positive percentage (0.066), negative percentage (0.059), and favorability (0.055) also contribute meaningfully but to a lesser degree, while different genre categories each contribute only around 1% or less. This confirms that while trailer engagement does matter, a film's budget and the size of its audience interest, as reflected by vote count, are the dominant drivers of box office revenue.

Random Forest Visualization:



Decision Tree Visualization:

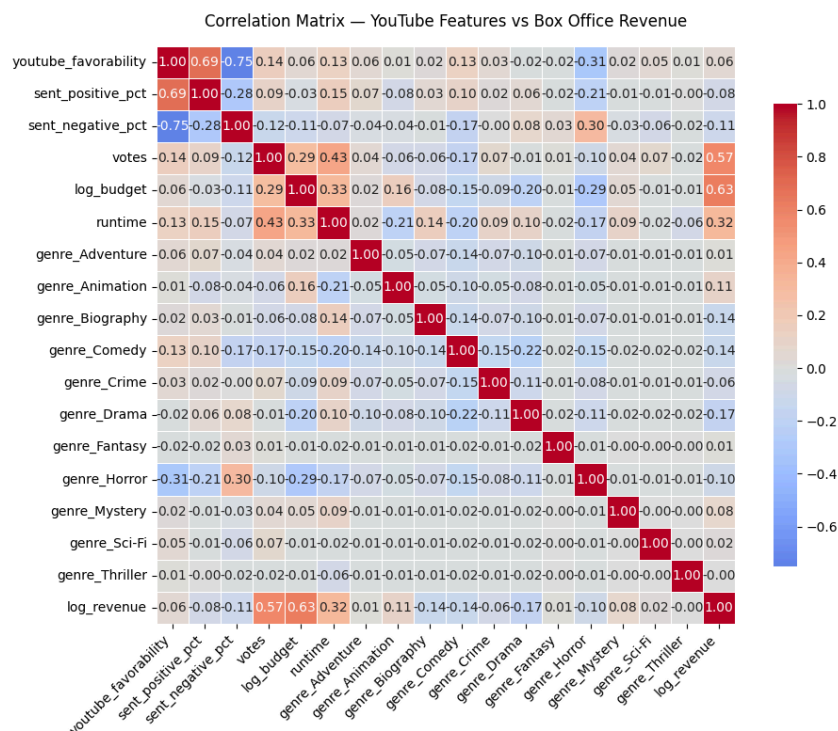


The Correlation Matrix

The correlation matrix was used after advanced modelling to give a simple overview of how each of the chosen features related to box office revenue. The target was simplified by using **log_revenue** instead of box office revenue because movie earnings are highly skewed. The missing values were handled through the cleaned **model_df**, ensuring that only movies with the required variables were included. Features that might cause data leakage or are not useful for prediction, like titles, links, and IDs, were removed. Due to time and scope constraints, the analysis is restricted to key variables in the set **ALL_FEATURES**, keeping the analysis simple and easy to understand. The tradeoff for the correlation matrix is easy to interpret, but it is limited to showing simple linear relationships.

To run the correlation matrix in *Google Colab*, the project requires the cleaned, merged movie data used to create **model_df**, and the notebook containing the analysis code. The libraries that are required for this project are **pandas**, **matplotlib**, and **seaborn**. This includes the cell to create **corr_matrix** with **model_df[ALL_FEATURES + [TARGET]]**, the cell where the correlation table is printed, and where the heatmap is created. The output appears immediately below the cells in Colab in the form of a printed correlation matrix and heatmap.

The correlation matrix shows that **log_budget** (0.63) and **votes** (0.57) have the highest positive correlations with **log_revenue**, followed by a smaller positive correlation between **runtime** (0.32) and **log_revenue**. The YouTube trailer metrics, including **youtube_favorability** (-0.06), **sent_positive_pct** (-0.08), and **sent_negative_pct** (-0.11), show low correlations with **log_revenue**. Many of these genre variables have low correlations, but **genre_Drama** has the strongest negative correlation of -0.17. This suggests that box office performance in this dataset is more closely related to budget and audience activity, and less to trailer sentiment or genre. The correlation matrix does not measure model performance directly, but it is useful since it indicates the most important variables and helps to verify the trends seen in the model's results.



Model Comparison

To compare the four models, we looked at R^2 , MAE, and RMSE. Random Forest performed best overall, with the highest R^2 and the lowest error scores. Full Linear Regression came next, followed by Decision Tree, while the Baseline Linear Regression using only YouTube features performed worst. One important result is the improvement from the Baseline to the Full Linear Regression: adding movie-level control variables such as budget, runtime, and genre substantially improved performance. This suggests that YouTube metrics provide some predictive value, but they are not strong enough alone to explain box office revenue. The fact that Random Forest performed best also suggests that the relationship between the predictors and revenue is not purely linear. Overall, the results show that YouTube trailer engagement, especially engagement volume, contains useful signals, but it works best when combined with broader movie characteristics.

	Model	R^2	MAE	RMSE
3	Random Forest	0.5730	0.4827	0.5777
1	Full LR (YouTube + Controls)	0.5260	0.5026	0.6087
2	Decision Tree	0.4319	0.5487	0.6664
0	Baseline LR (YouTube Only)	0.3324	0.5755	0.7224

Conclusion

This project aimed to answer whether YouTube trailer engagement metrics can predict a movie's box office revenue. The results show that while trailer engagement does have predictive value, it is not sufficient on its own, as the baseline linear regression model explained only about one-third of the variation in revenue. The Random Forest model performed best, achieving the highest R^2 of 0.573 and the lowest prediction error, indicating that non-linear relationships are important in capturing the complexity of box office performance. Across all models, log budget and YouTube vote count consistently emerged as the strongest predictors, highlighting the importance of both financial investment and audience interest. Overall, the findings suggest that YouTube engagement metrics are a useful but incomplete indicator and are most effective when combined with other key variables in a more advanced modelling approach.